



Professional Data Analysis & Intelligent Reporting

Professional Data Analysis Report

Comprehensive Analysis & Insights

Report Generated:	December 23, 2025
Platform:	Insightify Professional Analytics
Engine Version:	3.0 (Enhanced AI Analytics)
Report Status:	Confidential

Generated by Insightify™ - AI-Powered Data Analysis Platform
Transforming Data into Actionable Insights

■ Executive Summary

Report Overview:

This comprehensive data analysis report presents key insights and actionable metrics from the provided dataset. Our analysis covers **937** records across **4** columns, utilizing advanced statistical methods and visualization techniques.

Dataset Composition:

- Numeric Variables: **3** columns
- Categorical Variables: **1** columns
- Temporal Variables: **0** columns
- Data Completeness: **0** missing values detected
- Data Uniqueness: **0** duplicate rows identified
- Memory Footprint: **0.08 MB**

■ Key Insights & Findings

Our analysis has uncovered the following significant patterns:

1. venda shows significant right-skewed distribution (skewness: 1.43)
2. estoque shows significant right-skewed distribution (skewness: 1.76)
3. data has very high cardinality (937 unique values)

■ Numeric Analysis

Column	Mean	Median	Std Dev	Min	Max	IQR
venda	90.53	76.00	80.68	0.00	542.00	94.00
estoque	1608.26	1348.00	1356.69	0.00	7228.00	1170.00
preco	1.59	1.39	0.53	0.00	2.98	0.60

■ Categorical Analysis

Category: data

Unique Values: 937 | Most Frequent: '2014-01-01' (0.1% of data)

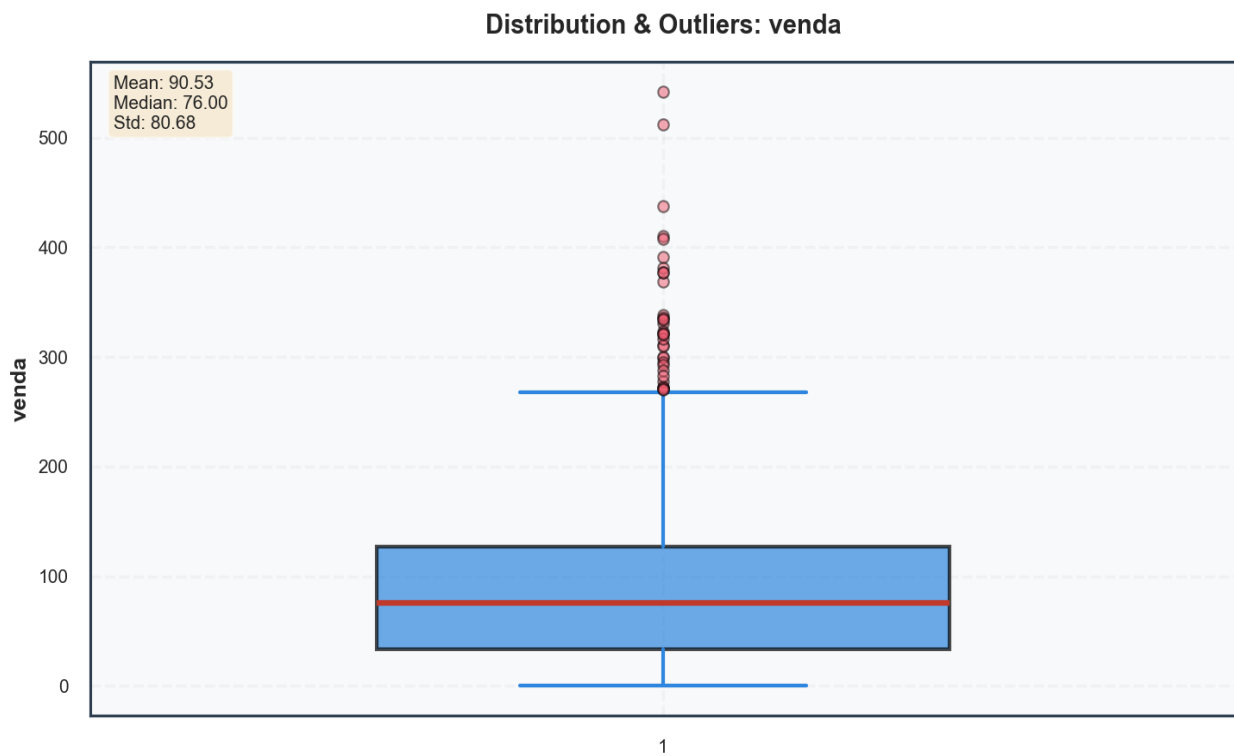
Value	Count	Percentage
2014-01-01	1	10.0%
2014-01-02	1	10.0%
2014-01-03	1	10.0%
2014-01-04	1	10.0%
2014-01-05	1	10.0%
2014-01-06	1	10.0%
2014-01-07	1	10.0%
2014-01-08	1	10.0%
2014-01-09	1	10.0%
2014-01-10	1	10.0%

■ Correlation Analysis

No significant correlations detected in the dataset.

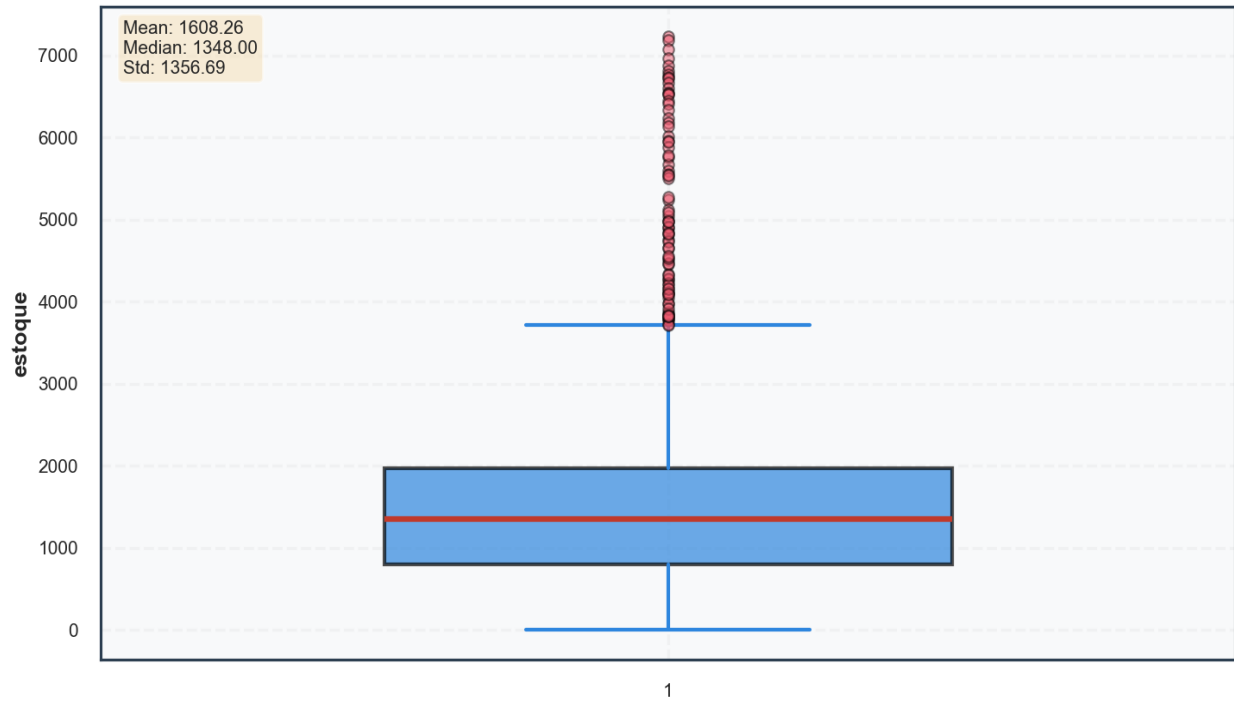
■ Data Visualizations

Boxplot Venda

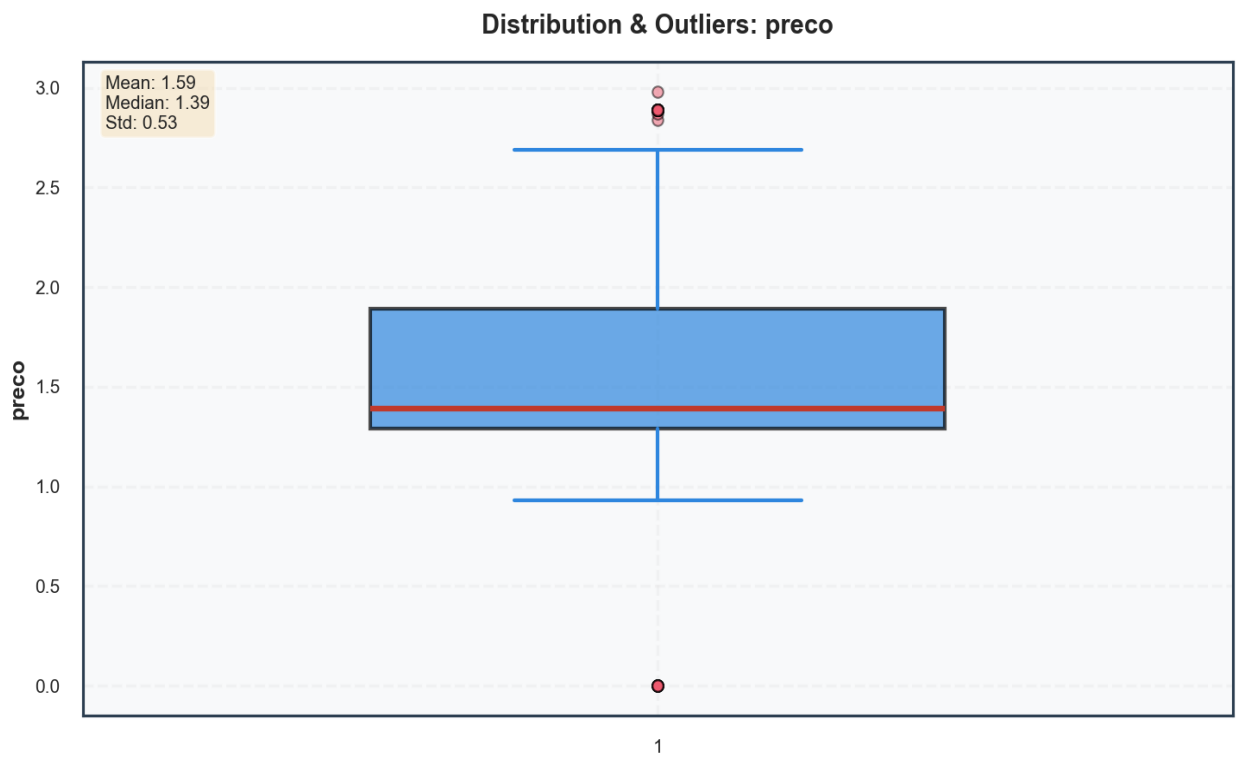


Boxplot Estoque

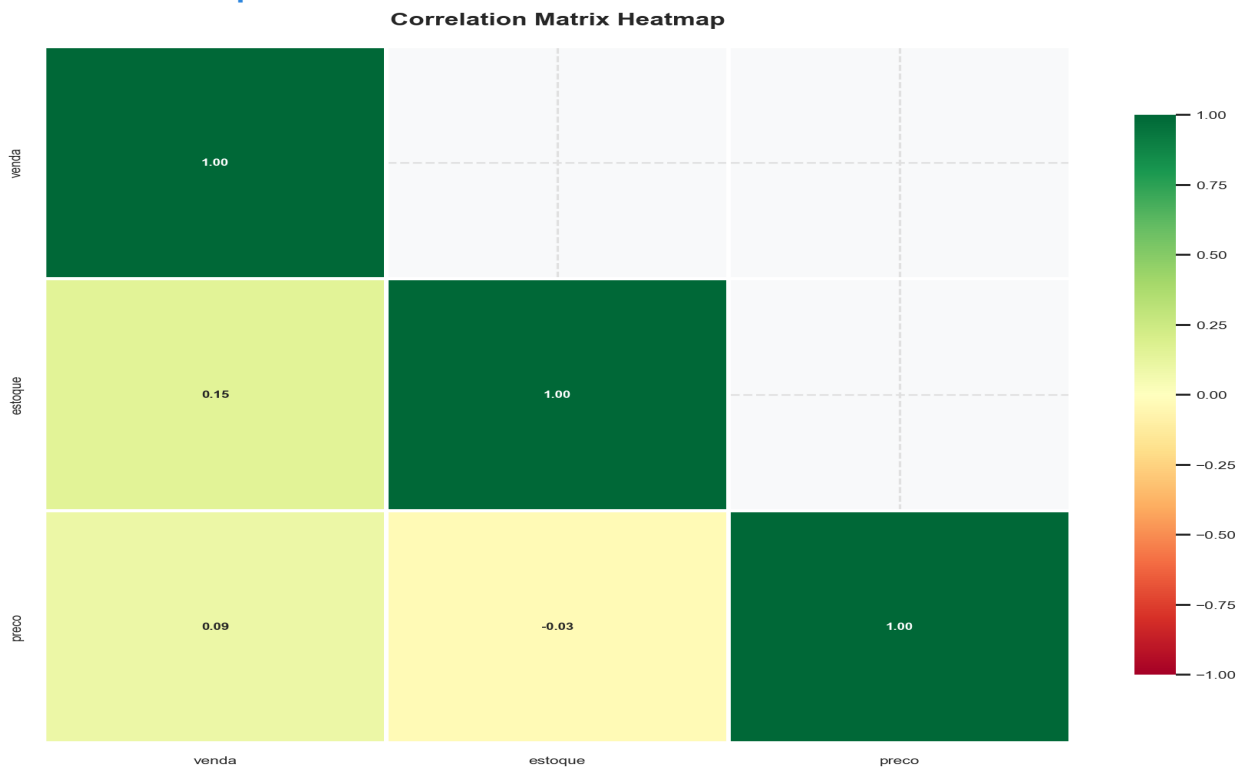
Distribution & Outliers: estoque



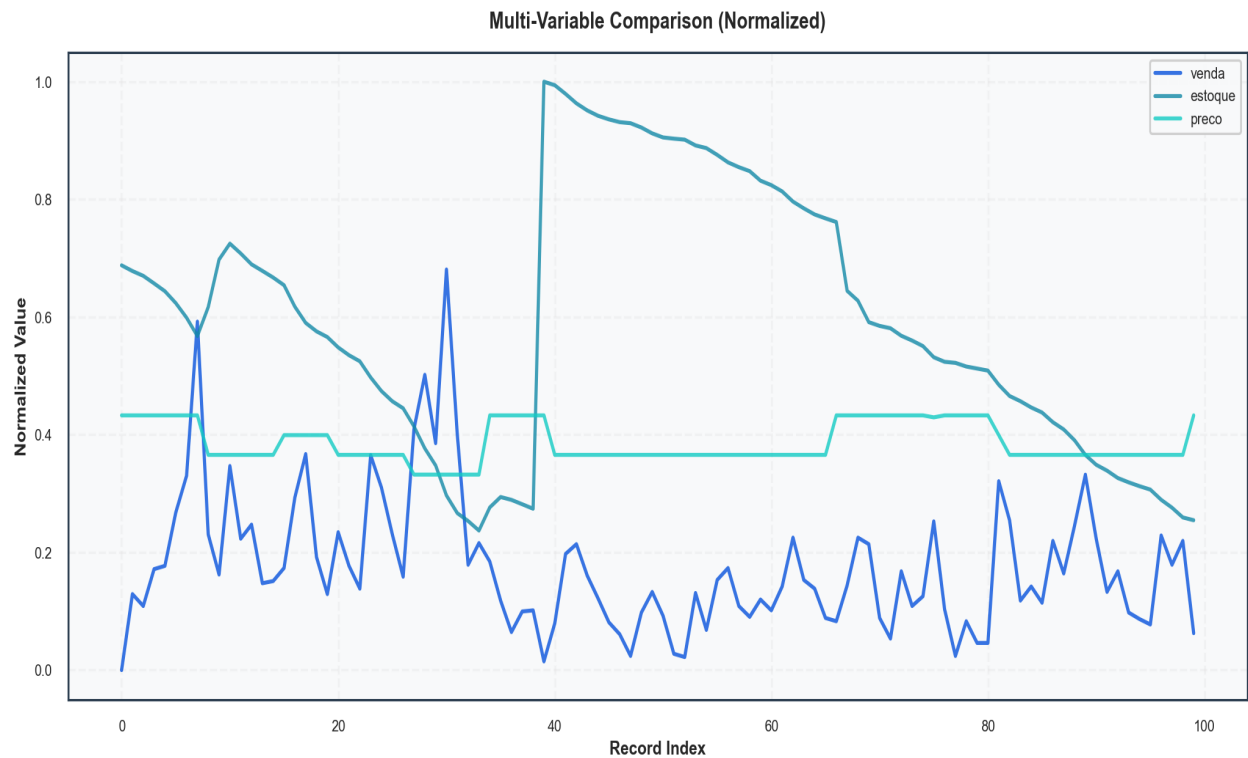
Boxplot Preco



Correlation Heatmap



Multiline Comparison



■ Conclusions & Recommendations

Key Findings:

1. Data Quality Assessment:

The dataset has undergone comprehensive quality checks. Review the Executive Summary for critical insights regarding data completeness, duplicate records, and overall integrity. Address any missing values or duplicates identified before proceeding with predictive modeling.

2. Statistical Distribution Analysis:

Our numeric analysis reveals the central tendencies, variability, and distribution patterns across all quantitative variables. Box plots highlight potential outliers that may require further investigation or treatment depending on your analytical objectives.

3. Correlation & Relationship Discovery:

The correlation analysis has identified statistically significant relationships between variables. Strong positive or negative correlations ($|r| > 0.7$) indicate potential predictive relationships that could be leveraged in machine learning models or business intelligence applications.

Actionable Recommendations:

✓ **Investigate High-Correlation Pairs:** Variables with strong correlations warrant deeper analysis to determine causality versus mere association. Consider domain expertise when interpreting these relationships.

✓ **Address Data Quality Issues:** Implement data cleaning protocols for missing values and duplicates. Consider imputation strategies or removal based on the nature and extent of missing data.

✓ **Leverage Visualizations for Stakeholder Communication:** The generated charts provide clear, professional representations of your data suitable for executive presentations and decision-making forums.

✓ **Monitor Categorical Distributions:** Highly concentrated categories or imbalanced classes may require special handling in predictive models (e.g., SMOTE for oversampling, stratified sampling).

✓ **Temporal Pattern Exploration:** If time-series trends are present, consider seasonal decomposition and forecasting techniques to predict future patterns.

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